a Puzzle (natural language)

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hobby: cooking, tennis, chess, painting, golf
job: pilot, farmer, baker, lawyer, teacher
color: yellow, white, purple, red, green
nationality: Spaniard, Brit, German, Swede, Dane
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1. The person who enjoys tennis and the farmer are different people.
2. The person who enjoys painting lives directly to the left of the person who enjoys golf.
3. The Swede and the person in the white house live two houses apart.
4. The baker lives in the house just left of the one where the Swede lives.
5. The teacher and the occupant of the yellow house are neighbours.
6. The Dane lives in the green house.
7. The cooking enthusiast lives in the red house.
8. The baker lives directly to the left of the person who enjoys chess.
9. Whoever lives in the white house lives in a lower-numbered house than the person who enjoys chess.
10. The lawyer and the teacher live two houses apart.
11. The teacher lives immediately to the right of the person who enjoys cooking.
12. The occupant of the red house and the Swede live in adjacent houses.
13. The Spaniard and the Swede live in adjacent houses.
14. The German lives in house 3.

b Two independent encodings

A and B are separate Sonnet calls; z3 proves (A xor B) unsatisfiable, i.e. A and B are equivalent.

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1  A: pos('tennis') != pos('farmer')
   B: (identical up to case)

2  A: pos('golf') == pos('painting') + 1
   B: pos('painting') + 1 == pos('golf')

3  A: Abs(pos('Swede') - pos('white')) == 2
   B: (identical up to case)

4  A: pos('baker') == pos('Swede') - 1
   B: pos('baker') + 1 == pos('swede')

5  A: Abs(pos('teacher') - pos('yellow')) == 1
   B: (identical up to case)

6  A: pos('Dane') == pos('green')
   B: (identical up to case)

7  A: pos('cooking') == pos('red')
   B: (identical up to case)

8  A: pos('chess') == pos('baker') + 1
   B: pos('baker') + 1 == pos('chess')

9  A: pos('white') < pos('chess')
   B: (identical up to case)

10 A: Abs(pos('lawyer') - pos('teacher')) == 2
   B: (identical up to case)

11 A: pos('teacher') == pos('cooking') + 1
   B: (identical up to case)

12 A: Abs(pos('red') - pos('Swede')) == 1
   B: (identical up to case)

13 A: Abs(pos('Spaniard') - pos('Swede')) == 1
   B: (identical up to case)

14 A: pos('German') == 3
   B: (identical up to case)
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c Forced deductions (ladder L3)

facts in the order the ladder engine derives them; L = highest rule needed so far; {..} = minimal set of clues that forces the fact

1	german = house 3	L1	{14}
2	cooking = house 3	L3	{4,5,7,8,9,11,12,14}
3	tennis = house 5	L3	{2,3,4,5,7,8,9,11,12,14}
4	chess = house 4	L3	{3,4,5,7,8,9,11,12,14}
5	painting = house 1	L3	{2,3,4,5,7,8,9,11,12,14}
6	golf = house 2	L3	{2,3,4,5,7,8,9,11,12,14}
7	pilot = house 5	L3	{1,2,3,4,5,7,8,9,10,11,12,14}
8	farmer = house 1	L3	{1,2,3,4,5,7,8,9,10,11,12,14}
9	baker = house 3	L3	{3,4,5,7,8,9,11,12,14}
10	lawyer = house 2	L3	{4,5,7,8,9,10,11,12,14}
11	teacher = house 4	L3	{4,5,7,8,9,11,12,14}
12	yellow = house 5	L3	{4,5,7,8,9,11,12,14}
13	white = house 2	L3	{3,4,5,7,8,9,11,12,14}
14	purple = house 4	L3	{3,4,5,6,7,8,9,11,12,14}
15	red = house 3	L3	{4,5,7,8,9,11,12,14}
16	green = house 1	L3	{3,4,5,6,7,8,9,11,12,14}
17	spaniard = house 5	L3	{3,4,5,7,8,9,11,12,13,14}
18	brit = house 2	L3	{3,4,5,6,7,8,9,11,13,14}
19	swede = house 4	L3	{3,4,5,7,8,9,11,12,14}
20	dane = house 1	L3	{3,4,5,6,7,8,9,11,13,14}